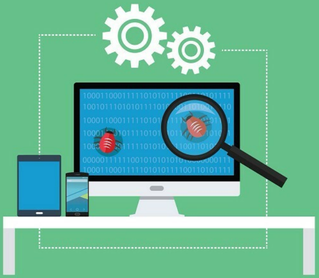
OpenStack:
Testing Tempest and REST API



Large pools of compute, storage and networking resources exist in data centres and in enterprises. To manage them, the ideal tool would be OpenStack, which gives administrative control via a dashboard. In this article, we will explore the configuration of DevStack and Tempest, looking deeper into the challenges and analysing Tempest test results across different OSs and hardware platforms.

In today's fast-paced world, resource sharing and resource optimisation are two big game changers, as these not only simplify infrastructure planning but also hold promise in the area of cloud computing.

Cloud computing has several advantages. It helps to save on capital and operational expenditure, and is easy to use. Cloud computing involves a big pool of hardware systems connected in private or public networks, to provide scalable infrastructure for application, data and file storage.

OpenStack

OpenStack is a cloud operating system that controls large pools of compute, storage and networking resources through a data centre used for developing private and public clouds. Being an open source platform, it gives administrators control through a dashboard (usually Horizon or a custom-built one using REST APIs), while empowering users to provision resources through a Web interface.

The OpenStack Foundation, a non-profit organisation, oversees both development and community-building around these projects, and is backed by some of the biggest companies in software development and hosting.

Generally speaking, OpenStack fits between the PaaS and IaaS layers, though not compulsorily so. It lets users deploy virtual machines (VMs) and other instances, which handle different tasks that manage a cloud environment, on-the-fly. It makes horizontal scaling easier, which means that tasks which benefit from running concurrently can easily serve more or fewer users on-the-fly by just spinning up more instances as and when required.

OpenStack was initially incubated and developed by NASA, CERN and Rackspace. It is backed by thousands of developers around the world, who are ready to help build a strong, robust and secure environment. OpenStack has well integrated features and components to carry out all cloud related activities like provisioning, orchestration, pricing, billing, etc.

This free and open source cloud computing platform could reach an estimated market size of US$ 1.7 billion by the end of 2016, according to a report from the analyst firm 451 Research.

A recent announcement at *OpenStack.org* stated: "OpenStack is enterprise ready." In fact, OpenStack is revolutionising the way we approach computing, storage and networking.